

FREEDOM INTERNATIONAL SCHOOL
WORKSHEET-MCQ-SOLUTIONS
CLASS XI
PHYSICS
MECHANICAL PROPERTIES OF SOLIDS

1.	A stretched rubber has (a) increased kinetic energy (b) increased potential energy (c) decreased kinetic energy (d) decreased potential energy Ans: (b) increased potential energy	1
2.	Which of the following affects the elasticity of a substance? (a) hammering and annealing (b) change in temperature (c) impurity in substance (d) all of these (d) All of these	1
3.	The diameter of a brass rod is 4 mm. Young's modulus of brass is $9 \times 10^{10} \text{ N/m}^2$. The force required to stretch 0.1% of its length is (a) $360 \pi \text{ N}$ (b) 36 N (c) $36 \pi \times 10^5 \text{ N}$ (d) $144 \pi \times 10^3 \text{ N}$ Ans: (a) $360 \pi \text{ N}$ $r = 2 \times 10^{-3} \text{ m}, Y = 9 \times 10^{10} \text{ N/m}^2,$ $l = 0.1\% L \Rightarrow \frac{l}{L} = 0.001$ As $Y = \frac{F L}{A l}$ $\therefore F = YA \frac{l}{L}$ $= 9 \times 10^{10} \times \pi (2 \times 10^{-3})^2 \times 0.001$ $= 360 \pi \text{ N}$	1
4.	A sphere of radius 3 cm is subjected to a pressure of 100 atm. Its volume decreases by 0.3 c.c. What will its bulk modulus be? (a) $4\pi \times 10^5 \text{ atm}$ (b) $4\pi \times 3 \times 10^3 \text{ atm}$ (c) $4\pi \times 10^6 \text{ atm}$ (d) $4\pi \times 10^8 \text{ atm}$ Ans: (b) $4\pi \times 3 \times 10^3 \text{ atm}$	1

	Given, $r = 3\text{cm}$, $\Delta V = 0.3$ Bulk modulus, $B = \frac{pV}{\Delta V} = \frac{100 \times \frac{4}{3}\pi r^3}{0.3}$ $= 4\pi \times 3 \times 10^3 \text{atm}$	
5.	The Young's modulus of a wire of length L and radius r is Y newton per square metre. If the length is reduced to $L/2$ and radius $r/2$, its Young's modulus will be (a) $Y/2$ (b) Y (c) $2Y$ (d) $4Y$ (b) Y It depends on the nature of the material	1
6.	A wire whose cross-sectional area is 2 mm^2 is stretched by 0.1 mm by a certain load, and if a similar wire of triple the area of cross-section is stretched by the same load, then the elongation of the second wire would be (a) 3.3 mm (b) 0.033 mm (c) 0.33 mm (d) 0.0033 mm Ans: (b) 0.033 mm $\Delta l = \frac{F}{A} \cdot \frac{l}{Y}$ For same F, l, Y $\Delta l \propto \frac{1}{A}$ $\therefore \frac{\Delta l_2}{\Delta l_1} = \frac{A_1}{A_2} = \frac{2\text{ mm}^2}{6\text{ mm}^2}$ $\therefore \Delta l_2 = \frac{1}{3} \times 0.1\text{ mm} = 0.033\text{ mm}$	1
7.	The work done per unit volume in deforming a body is given by (a) stress x strain (b) $\frac{1}{2}$ (stress x strain) (c) stress/strain (d) strain/stress Ans: (b) $\frac{1}{2}$ (stress x strain)	1
8.	A wire fixed at the upper end stretches by length l by applying a force F. The work done in stretching is (a) $F/2l$ (b) Fl (c) $2Fl$ (d) $Fl/2$ Ans: (d) $Fl/2$ $\text{Young's modulus } Y = \frac{FL}{Al} \quad \dots\dots (i)$ $\therefore F = \frac{YAl}{L}$ or $dW = F dl = \frac{YAl(dl)}{L}$ or $\int dW = \frac{YA}{L} \int_0^l l dl = \frac{YAl^2}{2L}$ or Workdone $= \frac{YAl^2}{2L} \quad \dots\dots (ii)$ From (i) and (ii) $\text{Workdone} = \frac{Fl}{2}$	1

9.	If longitudinal strain for a wire is 0.03 and its Poisson ratio is 0.5, then its lateral strain is (a) 0.003 (b) 0.0075 (c) 0.015 (d) 0.4 Ans: (c) 0.015 Poisson's ratio = [(lateral strain) / (longitudinal strain)] $\therefore 0.5 = [(\text{lateral strain}) / (0.03)]$ $\therefore \text{lateral strain} = 0.015$	1
10.	The Young's modulus of steel is twice that of brass. Two wires of same length and of same area of cross section, one of steel and another of brass are suspended from the same roof. If we want the lower ends of the wires to be at the same level, then the weights added to the steel and brass wires must be in the ratio of (a) 1:1 (b) 1:2 (c) 2:1 (d) 4:1 Ans: (c) 2:1 Young's modulus, $Y = \frac{Fl}{A\Delta l}$ here A, l, and Δl is same for both material, so, $F \propto Y$ Thus, $\frac{F_s}{F_b} = \frac{Y_s}{Y_b} = \frac{2Y_b}{Y_b} = 2$	1